

Michael Ryan Zengel

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Objective

First-year Physics PhD student at MIT, focusing on ultrafast optical spectroscopy and light-driven dynamics in quantum materials. Experienced in THz emission spectroscopy, first-principles theory, and machine learning for materials discovery.

Education

Massachusetts Institute of Technology, Cambridge, MA, PhD in Physics Aug 2025 – Present

- National Science Foundation Graduate Research Fellow
- MIT Dean of Science Fellow

University of Alabama, Tuscaloosa, AL, BS in Physics and Mathematics Aug 2021 – May 2025

- Graduated *summa cum laude* | GPA: 4.0/4.0
- **Minor**: Randall Research Scholars Program; French

Research Experience

Heusler Material Research, UA Department of Physics – Tuscaloosa, AL May 2022 – May 2025

- Worked with Dr. Adam Hauser to predict the atomic structure and crystalline, magnetic, and conductive properties of 3 element Heusler alloys and check predictions against grown materials.
- Employed Quantum Espresso, a Density Functional Theory Software, on the High-Performance Computing Center to calculate lattice constants, formation energy, density of state around the Fermi energy, and the band structure of alloys.
- Developed training course and instructed new members of lab how to perform DFT calculations.

University of Michigan CERN REU, CERN – Geneva, Switzerland June 2024 – Aug 2024

- Participated in the CERN Summer Student Program.
- Worked with the Engagement Group of the Strategy and Executive Governance Division of the IT department (IT-GOV-ENG) and developed a Jupyter Lab Extension for Zenodo, a dataset sharing software, of which I will be an ongoing contributor.
- Attended a lecture series by leading CERN scientists discussing particle physics and accelerator design.

Ancient Plants and People Lab, UA Department of Anthropology – Tuscaloosa, AL March 2023 – May 2024

- Developed Python framework to dynamically organize and refine survey data concerning students' perceived food security on the University of Alabama's campus.
- Developed project which uses machine learning and convolutional neural networks to dynamically identify seed species based on shape with respect to morphometrics.

DAAD RISE Summer Research, Justus Liebig University – Gießen, Germany May 2023 – Aug 2023

- Designed and constructed an optical setup which generates terahertz frequency via photocurrent injection in semiconductors using a two-color excitation scheme.
- Analyzed the mobility and conductivity of Metal-organic chemical vapor deposition grown monolayers using THz time-domain spectroscopy.

Compact Muon Solenoid Research, UA Department of Physics – Tuscaloosa, AL Sept 2021 – July 2022

- Worked with Dr. Sergei Gleyzer to determine the efficacy of convolutional neural networks in classifying raw jet data from the Compact Muon Solenoid at the Large Hadron Collider at CERN.
- Generated and improved upon several convolutional neural networks.
- Researched the application of data augmentation techniques on the convergence of classification.

DFT Machine Learning Summer Research, Tulane University Physics Department June 2019 – Aug 2021

– New Orleans, LA

- Worked with Dr. Jianwei Sun to research the applications of machine learning in Density Functional Theory.
- Produced several 2D and convolutional neural networks to predict certain values with the SCAN method of calculation.
- Developed a script that exploited point group symmetry to automate the generation of unique grids around molecules to reduce redundancy within input files.

Publications

The 4-2 rule: The (Proposed) Answer to Heusler Alloys, Stability, and (Ordering) Everything Dec 2025

Ridwan Nahar, Riley Nold, Naomi Derksen, Ka Ming Law, Babajide Akintunde, **Michael Zengel**, Thomas Roden, Justin Lewis, Harshil Avlani, Sujan Budhathoki, Patrick LeClair, Adam Hauser

Submitted

Coherent Control of Photocurrents in Germanium for Gapless and Broadband Terahertz Emission Aug 2025

Daniel Anders, **Michael R. Zengel**, Rodrigo Sandoval Rodriguez, Kalle Bräumer, Sangam Chatterjee, Markus Stein 10.1007/s10762-025-01078-0, *Journal of Infrared, Millimeter, and Terahertz Waves*

Bilinear and Biquadratic Interlayer Exchange Coupling in Asymmetric Jul 2025

Co₆₅Fe₃₅/Ru/Co₆₅Fe₃₅ Heterostructures

Upama Karki, Bhuwan Nepal, Naomi Derksen, Anupama Karki, Riley Nold, **Michael Zengel**, Adam J. Hauser, Gary J Mankey, Tim Mewes, Jamileh Beik Mohammadi

10.1016/j.jmmm.2025.173315, *Journal of Magnetism and Magnetic Materials*

Atomic Ordering and Electronic Structure of A₂YSn Heusler Alloys: A First-Principles Study Apr 2025

Michael R. Zengel, Riley Nold, Thomas Roden, Ka Ming Law, Ridwan Nahar, Justin Lewis, Adam J. Hauser

10.1016/j.jallcom.2025.179270, *Journal of Alloys and Compounds*

Phase and d-d hybridization control via electron count for material property control in the X₂FeAl material class Mar 2024

Ka Ming Law, Ridwan Nahar, Riley Nold, **Michael Zengel**, Justin Lewis, Adam J. Hauser

10.1016/j.jmmm.2024.171932, *Journal of Magnetism and Magnetic Materials*

Dataset on Density Functional Theory investigation of Ternary Heusler Alloys Dec 2023

Ridwan Nahar, Ka Ming Law, Thomas Roden, **Michael Zengel**, Justin Lewis, Sujan Budhathoki, Riley Nold, Harshil Avlani, Babajide Akintunde, Naomi Derksen, Adam J. Hauser

10.1016/j.dib.2023.109971, *Data in Brief*

Presentations and Posters

Connecting Zenodo and Analysis Facilities: a JupyterLab Extension Middleware Nov 2024

Astronomical Data Analysis Software and Systems, University of Malta, Malta
(Virtual Poster Presentation)

Atomic Ordering and Electronic Structure of X₂YSn Heusler Alloys: A First-Principles Study Mar 2024

University of Alabama's Undergraduate Research and Creative Activity Conference, Tuscaloosa, AL, United States
(Poster Presentation)

Atomic Ordering and Electronic Structure of X₂YSn Heusler Alloys: A First-Principles Study Mar 2024

APS March Meeting 2024, Minneapolis, MN, United States.

(Oral Presentation)

Ab-Initio Investigation of Novel $X_2Y\text{Sn}$ Heusler Compounds

Dec 2023

RRS Live, Tuscaloosa, AL., United States

(Oral Presentation)

Creating Currents from Light: Terahertz Emission from Novel Materials

Jul 2023

DAAD RISE Heidelberg Meeting 2023, Heidelberg, Baden-Württemberg, Germany

(Oral Presentation)

First Principles Investigation of Atomic Ordering in Novel Heuslers

Apr 2023

University of Alabama's Undergraduate Research and Creative Activity Conference, Tuscaloosa, AL, United States

(Poster Presentation)

First Principles Investigation of Atomic Ordering in Novel Heuslers

Mar 2023

Joint Mississippi State University, University of Mississippi, and University of Alabama Physics Graduate Student Association Symposium, Starkville, MS, United States

(Poster Presentation)

Atomically Ordered Complex Materials via Control of Entropic Mixing

Dec 2022

RRS Live, Tuscaloosa, AL, United States

(Oral Presentation)

The Variance of the Angle of Incidence in the Double Slit Experiment

Mar 2020

Greater New Orleans Science and Engineering Fair & Louisiana Science and Engineering Fair, New Orleans, LA, United States

(Poster Presentation)

Using Machine Learning to De-orbitalize Correlation Model Selection in the SCAN Functional

Aug 2019

Tulane University PEP Summer Colloquium, New Orleans, LA, United States

(Oral Presentation)

Employment History

Crew Member, Dat Dog – New Orleans, LA

May 2025 – Aug 2025

- Worked as line cook, with experience grilling, frying, and dressing orders
- Experience with standard restaurant operating procedure

Head Computer Lab Manager, Randall Research Scholars – Tuscaloosa, AL

Jan 2022 – May 2025

- Performed routine maintenance and software checks of Randall Research computer lab and general honors computer lab
- Maintained on 3 active websites and multiple servers
- Managed Windows operations such as Active Directory, Hyper-V virtual machines, Imaging, et cetera
- Supervised and mentored a team of 8 of other undergraduate students

Skills

Experimental Techniques: XRR/XRD scanning, LASER maintenance, Ultrafast Spectroscopy including THz-TDS and THz emission spectroscopy, Sputter methods in material growth

Languages: English, French (Fluent), German (Working Proficiency)

Programming Languages: Python, Fortran95, Java, Javascript, HTML, CSS, MATLAB, C#

Operating Systems and Tools: Bash, jQuery, Tensorflow, Keras API, Machine Learning, Web Design and Management, Node.js, Next.js, Webserver Management, React.js Framework

Technology Management: Active Directory, MECM (SCCM), Hyper-V Manager

Software: Quantum Espresso, OriginLab, LABVIEW

Awards and Certifications

National Science Foundation Graduate Research Fellow	Fall 2025
Dean of Science Fellow (MIT)	Fall 2025
Fulbright Research Award (Declined)	Spring 2025
Catherine J. Randall Student Excellence Award	Spring 2025
Fulbright U.S. Student Program Semifinalist	Fall 2024
UA Churchill Scholarship Nominee	Fall 2024
UA Marshall Scholarship Nominee	Fall 2024
UA Rhodes Scholarship Nominee	Fall 2024
Henry Copeland Scholarship	Spring 2024
Randall Outstanding Undergraduate Research Award	Spring 2024
Barry M. Goldwater Scholarship Recipient	Spring 2024
Invited as Young Scientist to Lindau Nobel Laureate Meeting	Spring 2024
University of Alabama Goldwater Scholarship Nominee	Fall 2023
DAAD RISE Scholarship Recipient for Summer Research	Summer 2023
First Place in Physical Sciences at UA's URCA Conference	Spring 2023
Health Track Winner in the University of Alabama BIG IDEAs Competition	Spring 2023
Outstanding Research Presentation in RRS	Spring 2023
Best Poster at the MSU-UM-UA PGSA Physics Symposium	Spring 2023
University of Alabama Goldwater Scholarship Nominee	Fall 2022
Outstanding First-Year Physics Student	Spring 2022
Outstanding Freshman in the Randall Research Scholars Program	Spring 2022
UA President's List	Fall 2021 - May 2025
National Merit Scholar	Spring 2021
University of Rochester Xerox Book Award	Spring 2021
Industry Based Certificate in Javascript Functional Programming and Web Design Level 1 and 2	Fall 2018
Diplôme d'Etudes en Langue Française (DELF) A2	Spring 2015

Activities

The Crimson White (Contributing Writer)	Fall 2024 - Spring 2025
Phi Beta Kappa	Spring 2024 - Spring 2025
Pi Mu Epsilon	Spring 2024 - Spring 2025
Society of Physics Students (President)	Fall 2021 - Spring 2025
Journal of Science and Health at UA (Editor, Delegate to NUCSJ)	Fall 2021 - Spring 2025
Sigma Pi Sigma	Spring 2023 - Spring 2025
RRS Freshman Mentor	Fall 2023 - Spring 2025

RRS Selection Committee Member

Fall 2022 - Spring 2025

RRS Finalist Weekend Volunteer

Spring 2023, 2024